

Phase 1 - Week 4 - Lesson Day 3

Confidence Intervals & Significance

Today's Goal

Yesterday we learned how to test if a difference is real. Today we learn how to say "how sure" we are about a number we calculated from data, like an average.

- Understand what a confidence interval is
- Calculate a simple confidence interval
- Understand Type I and Type II errors

1. What Is a Confidence Interval?

Simple definition: A confidence interval is a range of numbers where we believe the TRUE answer probably sits. Instead of saying one exact number, we say a range, plus how sure we are.

Example: instead of saying "the average test score is exactly 75," we say "we are 95% confident the true average test score is between 72 and 78." This is more honest, because we only tested a small group of students, not everyone.

What does "95% confident" actually mean? If we repeated this same study 100 times, about 95 of those times, the true average would fall inside our calculated range. It does NOT mean "there is a 95% chance the true value is in this range" - that is a common mistake. It means our method is reliable 95% of the time.

2. Calculating a Confidence Interval

```
import numpy as np
from scipy import stats

scores = [70, 72, 68, 75, 71, 69, 74, 73]

mean = np.mean(scores)
sem = stats.sem(scores) # standard error - how much the average might vary

confidence_interval = stats.t.interval(0.95, len(scores)-1, loc=mean, scale=sem)

print("Average:", mean)
print("95% confidence interval:", confidence_interval)
```

You don't need to understand every detail of the formula. What matters is reading the result: a **narrow range** means we have a precise, confident estimate. A **wide range** means we are less sure - often because we have fewer data points.

3. Type I and Type II Errors

Simple definition: When we test a hypothesis, we can make two different kinds of mistakes.

Type I Error (false alarm): we say there IS a real difference, but actually there isn't. We rejected the null hypothesis by mistake.

Type II Error (missed finding): we say there is NO real difference, but actually there IS one. We failed to notice something that was really there.

Simple way to remember: Type I is like a smoke alarm going off when there's no fire. Type II is like a smoke alarm staying silent during a real fire.

	We say: no real difference	We say: there IS a difference
Truth: no real difference	Correct!	Type I Error (false alarm)
Truth: there IS a difference	Type II Error (missed it)	Correct!

Today's Practice

Open [day-3-confidence-intervals.ipynb](#). All answers are already solved with simple code and clear comments.

Practice Questions (4 total - all solved)

1. Calculate a confidence interval

Calculate the 95% confidence interval for a small group of made-up numbers (e.g. prices, scores, or ages).

2. Read the result

Write a comment explaining, in plain words, what your confidence interval from Question 1 means.

3. Bigger group, narrower range

Repeat the confidence interval calculation with a MUCH bigger group of numbers (at least 50), and compare how much narrower the range becomes.

4. Type I vs Type II

Write one short example (in a comment) of a real-life situation that would be a Type I error, and one that would be a Type II error.

Once you and Halimo finish reviewing today's practice, move to Lesson Day 4 (A/B Testing + Week 4 Mini Project).